

✓ 9. Non-collusive Oligopoly

In this section we will first present three models of duopoly, which is the limiting case of oligopoly. The common characteristic of these models is that they assume a certain pattern of reaction of competitors in each period and despite the fact that the 'expected' reaction does not in fact materialise, the firms continue to assume that the initial assumption holds. In other words, firms are assumed never to learn from past experience, which makes their behaviour at least naïve (if not stupid).

I. COURNOT'S DUOPOLY MODEL¹

The earliest duopoly model was developed in 1838 by the French economist Augustin Cournot.

The model may be presented in many ways. The original version is quite limited in that it makes the assumption that the duopolists have identical products and identical costs. Actually Cournot illustrated his model with the example of two firms each owning a spring of mineral water, which is produced at zero costs. We will present briefly this version, and then we will generalise its presentation by using the *reaction curves* approach.

Cournot assumed that there are two firms each owning a mineral well, and operating with zero costs. They sell their output in a market with a straight-line demand curve. Each firm acts on the assumption that its competitor will not change its output, and decides its own output so as to maximise profit.

Assume that firm *A* is the first to start producing and selling mineral water. It will produce quantity *A*, at price *P* where profits are at a maximum (figure 9.1), because at this point $MC = MR = 0$. The elasticity of market demand at this level of output is equal to unity and the total revenue of the firm is a maximum. With zero costs, maximum *R* implies maximum profits, Π . Now firm *B* assumes that *A* will keep its output fixed (at *OA*), and hence considers that its own demand curve is *CD'*. Clearly firm *B* will produce half the quantity *AD'*, because (under the Cournot assumption of fixed output of the rival) at this level (*AB*) of output (and at price *P'*) its revenue and profit is at a maximum. *B* produces half of the market which has not been supplied by *A*, that is, *B*'s output is $\frac{1}{2} (= \frac{1}{2} \cdot \frac{1}{2})$ of the total market.

Firm *A*, faced with this situation, assumes that *B* will retain his quantity constant in the next period. So he will produce one-half of the market which is not supplied by *B*. Since *B* covers one-quarter of the market, *A* will, in the next period, produce $\frac{1}{2}(1 - \frac{1}{4}) = \frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8}$ of the total market.

¹ A. Cournot, *Recherches sur les Principes Mathématiques de la Théorie des Richesses* (Paris, 1938). Translated by N. T. Bacon, *Researches into the Mathematical Principles of the Theory of Wealth* (Macmillan, 1927).

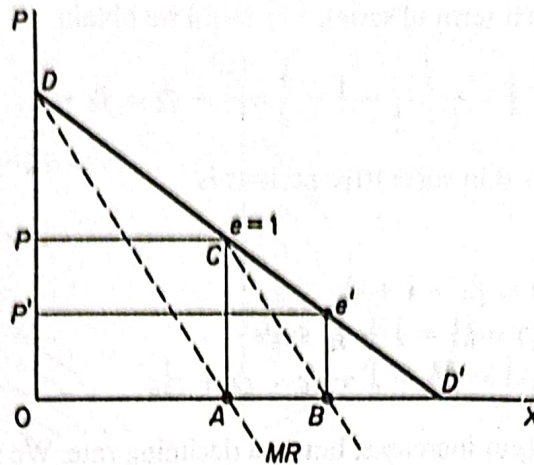


Figure 9.1

Firm B reacts on the Cournot assumption, and will produce one-half of the unsupplied section of the market, i.e. $\frac{1}{2}(1 - \frac{3}{8}) = \frac{5}{16}$.

In the third period firm A will continue to assume that B will not change its quantity, and thus will produce one-half of the remainder of the market, i.e. $\frac{1}{2}(1 - \frac{5}{16})$.

This action-reaction pattern continues, since firms have the naïve behaviour of never learning from past patterns of reaction of their rival. However, eventually an equilibrium will be reached in which each firm produces one-third of the total market. Together they cover two-thirds of the total market. Each firm maximises its profit in each period, but the industry profits are not maximised. That is, the firms would have higher joint profits if they recognised their interdependence, after their failure in forecasting the correct reaction of their rival. Recognition of their interdependence (or open collusion) would lead them to act as 'a monopolist,' producing one-half of the total market output, selling it at the profit-maximising price P, and sharing the market equally, that is, each producing one-quarter of the total market (instead of one-third).

The equilibrium of the Cournot firms may be obtained as follows:

1. The product of firm A in successive periods is

$$\text{period 1: } \frac{1}{2}$$

$$\text{period 2: } \frac{1}{2}(1 - \frac{1}{4}) = \frac{3}{8} = \frac{1}{2} - \frac{1}{8}$$

$$\text{period 3: } \frac{1}{2}(1 - \frac{5}{16}) = \frac{11}{32} = \frac{1}{2} - \frac{1}{8} - \frac{1}{32}$$

$$\text{period 4: } \frac{1}{2}(1 - \frac{43}{128}) = \frac{43}{128} = \frac{1}{2} - \frac{1}{8} - \frac{1}{32} - \frac{1}{128}$$

We observe that the output of A declines gradually. We may rewrite this expression as follows

$$\begin{aligned} \left[\begin{array}{l} \text{Product of A} \\ \text{in equilibrium} \end{array} \right] &= \frac{1}{2} - \frac{1}{8} - \frac{1}{32} - \frac{1}{128} \dots \\ &= \frac{1}{2} - \left[\frac{1}{8} + \frac{1}{8} \cdot \frac{1}{4} + \frac{1}{8} \cdot \left(\frac{1}{4}\right)^2 + \frac{1}{8} \left(\frac{1}{4}\right)^3 + \dots \right]. \end{aligned}$$

The expression in parentheses is a declining geometric progression with ratio $r = \frac{1}{4}$. Applying the summation formula for an infinite geometric series

$$\sum = \frac{a}{1-r}$$

(where $\sum$ = sum, a = first term of series, r = ratio) we obtain

$$\left[\begin{array}{l} \text{Product of } A \\ \text{in equilibrium} \end{array} \right] = \frac{1}{2} - \frac{\frac{1}{8}}{1 - \frac{1}{4}} = \frac{1}{2} - \frac{\frac{1}{8}}{\frac{3}{4}} = \frac{1}{2} - \frac{4}{24} = \frac{8}{24} = \frac{1}{3}$$

2. The product of firm B in successive periods is

$$\text{period 2: } \frac{1}{2}(\frac{1}{2}) = \frac{1}{4}$$

$$\text{period 3: } \frac{1}{2}(1 - \frac{3}{8}) = \frac{5}{16} = \frac{1}{4} + \frac{1}{16}$$

$$\text{period 3: } \frac{1}{2}(1 - \frac{11}{32}) = \frac{21}{64} = \frac{1}{4} + \frac{1}{16} + \frac{1}{64}$$

$$\text{period 4: } \frac{1}{2}(1 - \frac{43}{128}) = \frac{85}{256} = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \frac{1}{256}$$

We observe that B 's output increases, but at a declining rate. We may write

$$\left[\begin{array}{l} \text{Product of } B \\ \text{in equilibrium} \end{array} \right] = \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{4}(\frac{1}{4})^2 + \frac{1}{4}(\frac{1}{4})^3 + \dots$$

Applying the above expression for the summation of a declining geometric series we find

$$\left[\begin{array}{l} \text{Product of } B \\ \text{in equilibrium} \end{array} \right] = \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

Thus the Cournot solution is stable. Each firm supplies $\frac{1}{3}$ of the market, at a common price which is lower than the monopoly price, but above the pure competitive price (which is zero in the Cournot example of costless production). It can be shown that if there are three firms in the industry, each will produce one-quarter of the market and all of them together will supply $\frac{2}{3}$ ($= \frac{1}{4} \cdot 3$) of the entire market OD' . And, in general, if there are n firms in the industry each will provide $1/(n+1)$ of the market, and the industry output will be $n/(n+1) = 1/(n+1) \cdot n$. Clearly as more firms are assumed to exist in the industry, the higher the total quantity supplied and hence the lower the price. The larger the number of firms the closer is output and price to the competitive level.

Cournot's model leads to a stable equilibrium. However, his model may be criticised on several accounts:

The behavioural pattern of firms is naïve. Firms do not learn from past miscalculations of competitors' reactions.

Although the quantity produced by the competitors is at each stage assumed constant, a quantity competition emerges which drives P down, towards the competitive level.

The model can be extended to any number of firms. However, it is a 'closed' model, in that entry is not allowed; the number of firms that are assumed in the first period remains the same throughout the adjustment process.

The model does not say how long the adjustment period will be.

The assumption of costless production is unrealistic. However, it can be relaxed without impairing the validity of the model. This is done in the subsequent presentation of the model, based on the *reaction-curves* approach.

The reaction-curves approach is a more powerful method of analysis of oligopolistic markets, because it allows the relaxation of the assumption of identical costs and identical demands. This approach is based on *Stackelberg's indifference-curve analysis*, which introduces the concept of *isoprofit curves* of competitors. We will first establish the shape of the isoprofit curves for substitute commodities, and from these curves we will subsequently derive the reaction curves of the Cournot duopolists.

An isoprofit curve for firm A is the locus of points defined by different levels of output of A and his rival B , which yield to A the same level of profit (figure 9.2).

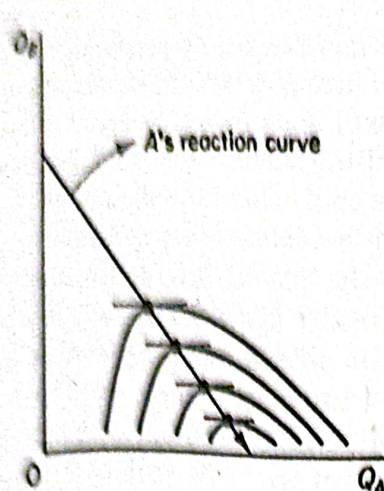


Figure 9.2 Isoprofit map of firm A

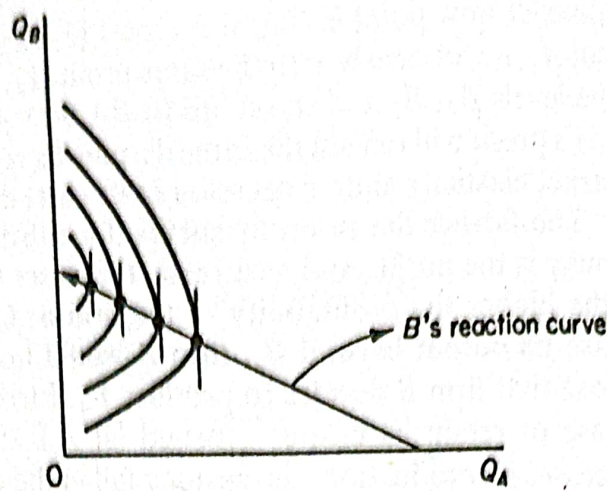


Figure 9.3 Isoprofit map of firm B

Similarly, an isoprofit curve for firm B is the locus of points of different levels of output of the two competitors which yield to B the same level of profit (figure 9.3).

From the above definitions it should be clear that the isoprofit curves are a type of indifference curves. There is a whole family of isoprofit curves for each firm which have the following properties.

1. Isoprofit curves for substitute commodities are concave to the axes along which we measure the output of the rival firms. For example, an isoprofit curve of firm A is concave to the horizontal axis Q_A . This shape shows how A can react to B's output decisions so as to retain a given level of profit. For example, consider the isoprofit curve Π_{A1} in figure 9.4.

Suppose that firm B decides to produce the level of output B_1 . A line parallel to the horizontal axis through B_1 intersects the isoprofit curve Π_{A1} at points h and g . This shows that given the output that B decides to produce, firm A will realise the profit Π_{A1} if it produces either of the two levels of output corresponding to points h and g , that is, either A_h or A_g . Assume that firm A decides to react by producing the higher level A_g . If now firm B increases its output (say at the level B_2), firm A must decrease its output (at A_f) if it wants to retain its profit at the same level (Π_{A1}). If firm A continued to produce A_g while B increased its production, the total quantity supplied in the market would depress the price, and hence the profit of firm A would decline. Up to a certain point (e in figure 9.4) firm A must react to increases in B's output by reducing its own production, otherwise the market price would fall and A's profit would decrease. As firm A reduces its output, its costs also change, but the net profit ($\Pi = R - C$) remains at the same level (Π_{A1}), because of market elasticity and/or decreasing costs arising from a better utilisation of A's plant.

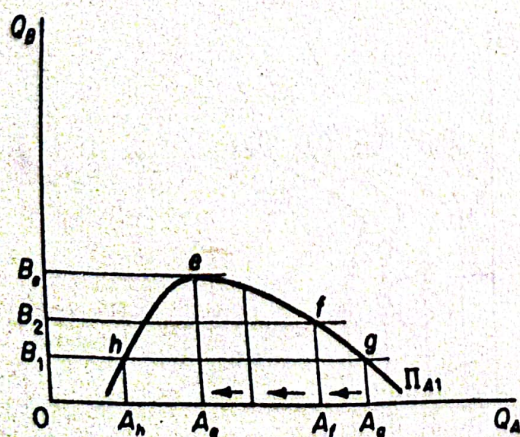


Figure 9.4

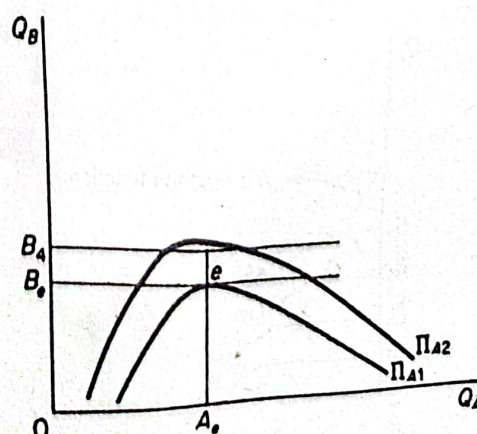


Figure 9.5

Consider now point h . If firm A reacts to B 's initial decision by producing the lower output A_h , it will clearly earn the same profit Π_{A1} . If firm B decides to increase its output (at the levels B_2 , B_3 and so on, up to B_e), firm A will react by increasing its output as well: A 's profit will remain the same despite the resulting fall in the market price, because of market elasticity and/or decrease in its costs due to a better utilisation of its plant.

2. The farther the isoprofit curves (for substitute commodities) lie from the axes, the lower is the profit. And vice versa, the closer to the quantity-axis an isoprofit curve lies, the higher the profitability of the firm is. Consider figure 9.5. If firm B were to increase its output beyond B_e , firm A would not be able to retain its level of profit. Suppose that firm B decides to produce B_4 . Firm A can react in three ways: increase, decrease or retain its output constant (at A_e). If A retains its output constant while B increases its production, the ensuing fall in the market price will result in a reduction in the revenue and in the profits of A , given its costs. If firm A were to increase output beyond A_e , its profit would fall because of inelasticity of demand and/or increasing costs. If firm A were to reduce output below A_e , its profit would fall because of elasticity of demand and/or increasing costs. Thus firm A will earn a lower level of profit, no matter what its reaction, if B increased its output beyond B_e . A line through B_4 parallel to the Q_A -axis lies above Π_{A1} , and will intersect (or will be tangent) to an isoprofit curve which represents a lower profit for firm A . In figure 9.5 the isoprofit curve Π_{A2} represents a lower profit than Π_{A1} . To summarise: for any given output that firm B may produce, there will be a unique level of output for firm A which maximises the latter's profit. This unique profit-maximising level of output will be determined by the point of tangency of the line through the given output of firm B and the lowest attainable isoprofit curve of firm A . In other words, the profit-maximising output of A (for any given quantity of B) is established at the highest point on the lowest attainable isoprofit curve of A .

3. For firm A , the highest points of successive isoprofit curves lie to the left of each other. If we join the highest points of the isoprofit curves we obtain firm A 's *reaction curve*. Thus, the reaction curve of firm A is the locus of points of highest profits that firm A can attain, given the level of output of rival B . It is called 'reaction curve' because it shows how firm A will determine its output as a reaction to B 's decision to produce a certain level of output. A 's reaction curve is shown in figure 9.6.

B 's isoprofit curves are concave to the Q_B axis. Their shape and position are determined by the same factors as the ones underlying firm A 's isoprofit curves. The highest point of the isoprofit curves of B lie to the right of each other as we move to curves further away from the Q_B axis. If we join these highest points we obtain B 's reaction function (figure 9.7). Each point of the reaction curve shows how much output B must produce in order to maximise its own profit, given the level of output of its rival.

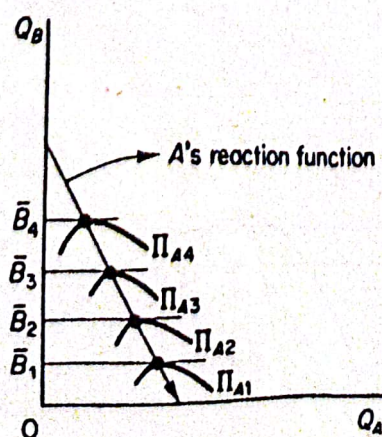


Figure 9.6

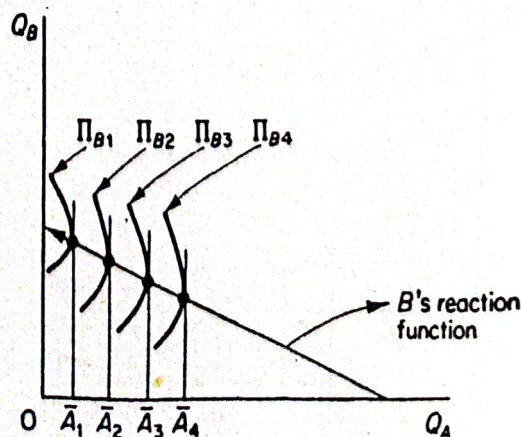


Figure 9.7

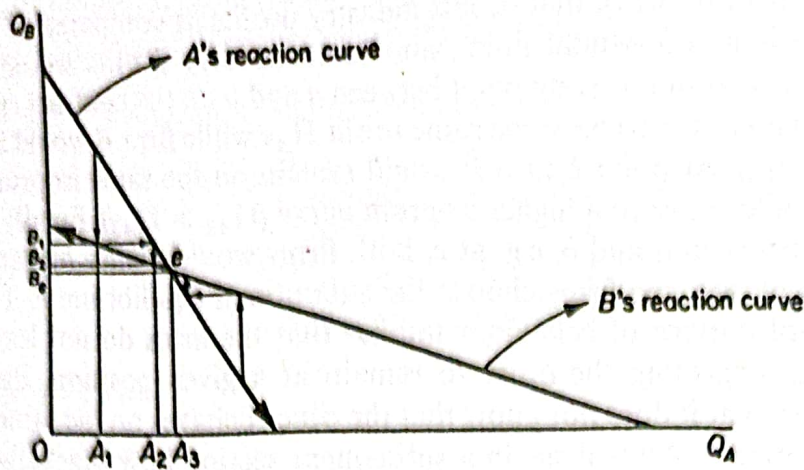
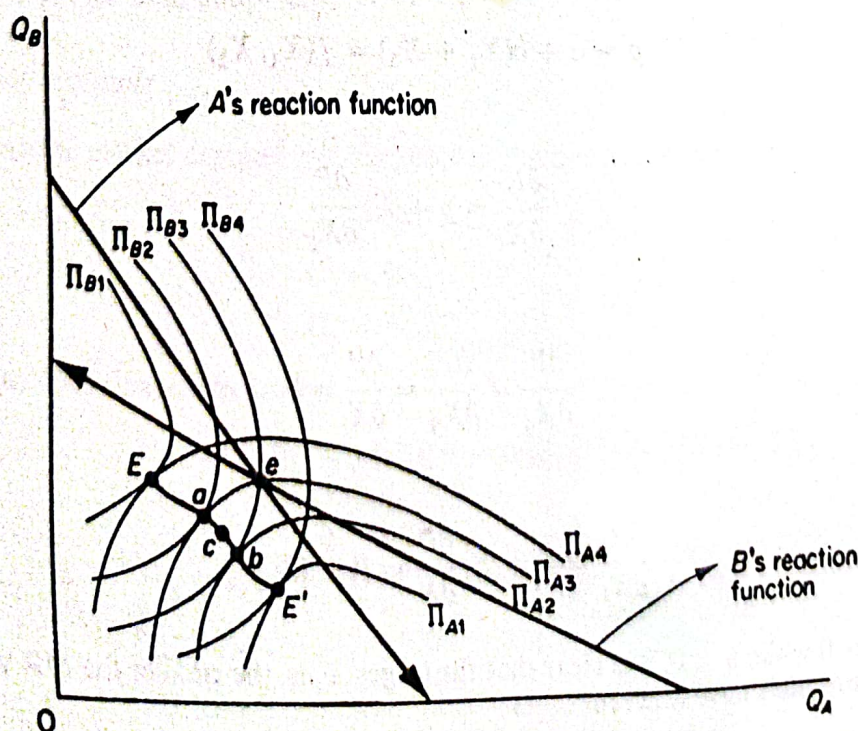


Figure 9.8

Cournot's equilibrium is determined by the intersection of the two reaction curves. It is a stable equilibrium, provided that A 's reaction curve is steeper than B 's reaction curve. (This condition is satisfied by the assumption we made that the highest points of successive isoprofit curves of A lie to the left of one another, while the highest points of B 's isoprofit curves lie to the right of each other.) To see that, let us examine the situation arising from A 's decision to produce quantity A_1 , lower than the equilibrium quantity A_e (figure 9.8). Firm B will react by producing B_1 , given the Cournot assumption that firm A will keep its quantity fixed at A_1 . However, A reacts by producing a higher quantity, of A_2 , on the assumption that B will stay at the level B_1 . Now firm B reacts by reducing its quantity at B_2 . This adjustment will continue until point e is reached. The same equilibrium would be reached if we started from a point to the right of e . Thus e is a stable equilibrium.

Note that at point e each firm maximises its own profit, but the industry (joint profit) is not maximised (figure 9.9). This is easily seen by a curve similar to Edgeworth's *contract curve* which traces points of tangency of the two firms' isoprofit curves. Points on the contract curve are optimal in the sense that points off this curve imply a lower

Figure 9.9. EE' is the 'Edgeworth's contract curve'

profit for one or both firms, that is, less industry profits as compared to points on the curve. Point e is a suboptimal point, and total industry profits would be higher if firms moved away from it on any point between a and b on the contract curve: at point a firm A would continue to have the same profit Π_{A3} while firm B would have a higher profit ($\Pi_{B2} > \Pi_{B3}$). At point b firm B would remain on the same isoprofit curve Π_{B3} while firm A would move to a higher isoprofit curve ($\Pi_{A2} > \Pi_{A3}$). Finally at any intermediate point between a and b , e.g. at c , both firms would realise higher profits. The question arises of why the firms choose the suboptimal equilibrium e . The answer is that the Cournot pattern of behaviour implies that the firms do not learn from past experience, each expecting the other to remain at a given position. Each firm acts independently, in that it does not know that the other behaves on the same assumption (behavioural pattern). We will see in a subsequent section how Stackelberg modified this model, by assuming that one or both of the duopolists may be sufficiently alert to recognise that his rival will make the Cournot assumption about his behaviour.

Mathematical derivation of the reaction curves:

A mathematical version of Cournot's model

Assume that the market demand facing the duopolists is

$$X = a + bP$$

or

$$P = a + bX \quad b < 0$$

Given that $X = X_1 + X_2$

$$\frac{\partial X}{\partial X_1} = \frac{\partial X}{\partial X_2} = 1$$

and the MRs of the duopolists need not be the same. Actually if the duopolists are of unequal size the one with the larger output will have the smaller MR.

Proof:

$$R_i = pX_i$$

$$p = a + b(X_1 + X_2) = f(X_1, X_2)$$

Thus

$$\frac{\partial R_i}{\partial X_i} = p + X_i \frac{\partial P}{\partial X_i}$$

But

$$\frac{\partial P}{\partial X_1} = \frac{\partial P}{\partial X_2} = \frac{\partial P}{\partial X} = b$$

Therefore

$$\frac{\partial R_i}{\partial X_i} = P + X_i \frac{\partial P}{\partial X} = P + (X_i)(b)$$

Given that $P > 0$ while $b < 0$, it is clear that the larger X_i is, the smaller the MR will be.
The two duopolists have different costs

$$C_1 = f_1(X_1) \quad \text{and} \quad C_2 = f_2(X_2)$$

Note that

$$MR_1 = \frac{\partial R_1}{\partial X_1} = \frac{\partial(PX_1)}{\partial X_1} = P + X_1 \frac{\partial P}{\partial X}$$

$$MR_1 = 45 + 80(-0.5)$$

$$MR_1 = 5$$

while $MR_2 = 45 + 30(-0.5)$

$$MR_2 = 30$$

That is, the firm with the larger output has the smaller marginal revenue. The profits of the duopolists are

$$\Pi_1 = PX_1 - C_1$$

$$\Pi_1 = (45)(80) - 5(80) = 3200$$

and

$$\Pi_2 = PX_2 - C_2$$

$$\Pi_2 = (45)(30) - 0.25(30)^2 = 900$$

The second-order condition is satisfied for both duopolists:

$$\left. \begin{aligned} \frac{\partial \Pi_1}{\partial X_1} &= 95 - X_1 - 0.5X_2 \\ \frac{\partial \Pi_2}{\partial X_2} &= 100 - 2X_2 - 0.5X_1 \\ \frac{\partial^2 \Pi_1}{\partial X_1^2} &= -1 < 0 \\ \frac{\partial^2 \Pi_2}{\partial X_2^2} &= -2 < 0 \end{aligned} \right\}$$

II. BERTRAND'S DUOPOLY MODEL

Bertrand¹ developed his duopoly model in 1883. His model differs from Cournot's in that he assumes that each firm expects that the rival will keep its price constant, irrespective of its own decision about pricing. Thus each firm is faced by the same *market demand*, and aims at the maximisation of its own profit on the assumption that the price of the competitor will remain constant.

The model may be presented with the analytical tools of the reaction functions of the duopolists. In Bertrand's model the reaction curves are derived from isoprofit maps which are convex to the axes, on which we now measure the prices of the duopolists. Each isoprofit curve for firm *A* shows the same level of profit which would accrue to *A* from various levels of prices charged by this firm and its rival. The isoprofit curve for *A* is convex to its price axis (P_A). This shape shows the fact that firm *A* must lower its price up to a certain level (point *e* in figure 9.11) to meet the cutting of price of its competitor, in order to maintain the level of its profits at Π_{A2} . However, after that price level has been reached and if *B* continues to cut its price, firm *A* will be unable to retain its profits, even if it keeps its own price unchanged (at P_{Ae}). If, for example, firm *B* cuts its price at P_B , firm *A* will find itself at a lower isoprofit curve (Π_{A1}) which shows lower profits. The reduction of profits of *A* is due to the fall in price, and the increase in output beyond the optimal level of utilisation of the plant with the consequent increase in costs. Clearly the lower the isoprofit curve, the lower the level of profits.

¹ See J. Bertrand, 'Théorie Mathématique de la Richesse Sociale', *Journal des Savants* (Paris, 1883) pp. 499-508.

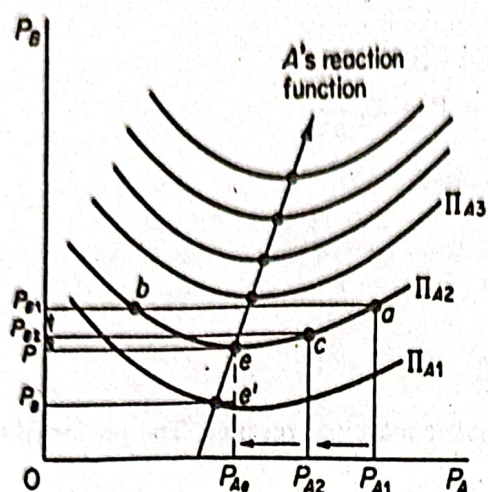


Figure 9.11

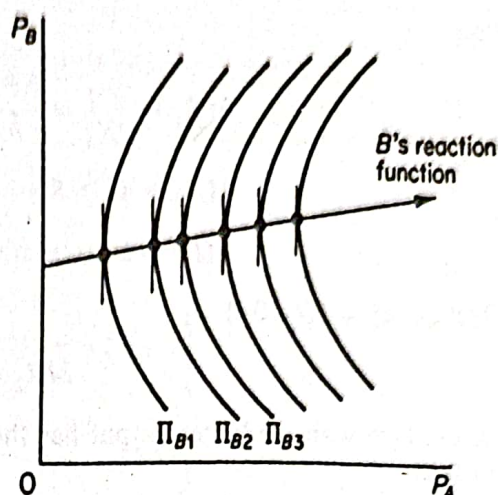


Figure 9.12

To summarise: for any price charged by firm *B* there will be a unique price of firm *A* which maximises the latter's profit. This unique profit-maximising price is determined at the lowest point on the highest attainable isoprofit curve of *A*. The minimum points of the isoprofit curves lie to the right of each other, reflecting the fact that as firm *A* moves to a higher level of profit, it gains some of the customers of *B* when the latter increases its price, even if *A* also raises its price. If we join the lowest points of the successive isoprofit curves we obtain the reaction curve (or conjectural variation) of firm *A*: this is the locus of points of maximum profits that *A* can attain by charging a certain price, given the price of its rival.

The reaction curve of firm *B* may be derived in a similar way, by joining the lowest points of its isoprofit curves (figure 9.12).

Bertrand's model leads to a stable equilibrium, defined by the point of intersection of the two reaction curves (figure 9.13). Point *e* denotes a stable equilibrium, since any departure from it sets in motion forces which will lead back to point *e* at which the price charged by *A* and *B* are P_{Ae} and P_{Be} respectively. For example, if firm *A* charges a lower price P_{A1} , firm *B* will charge P_{B1} , because on the Bertrand assumption, this price will maximise *B*'s profit (given P_{A1}). Firm *A* will react to this decision of its rival by charging

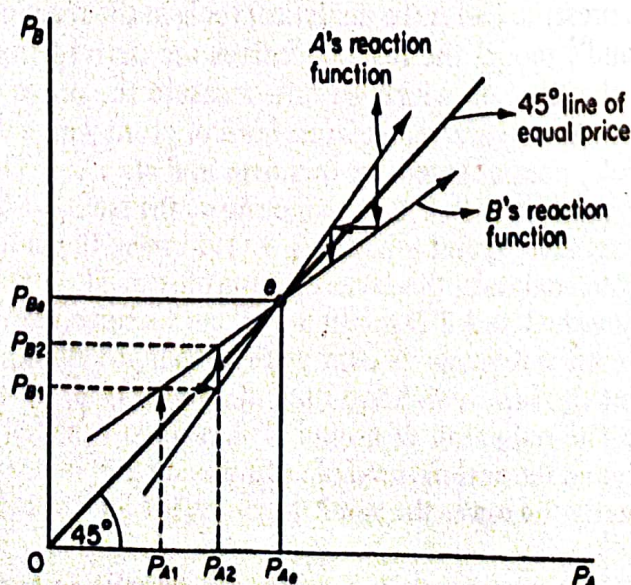


Figure 9.13

a higher price P_{A2} . Firm B will react by increasing its price, and so on, until point e is reached, when the market will be in equilibrium. The same equilibrium will be reached if firms started by charging a price higher than P_{Ae} or P_{Be} : a competitive price cut would take place which would drive both prices down to their equilibrium level P_{Ae} and P_{Be} .

Note that Bertrand's model does not lead to the maximisation of the industry (joint) profit, due to the fact that firms behave naïvely, by always assuming that their rival will keep its price fixed, and they never learn from past experience which showed that the rival did not in fact keep its price constant. The industry profit could be increased if firms recognised their past mistakes and abandoned the Bertrand pattern of behaviour (figure 9.14). If firms moved on any point between c and d on the *Edgeworth contract curve* (which is the locus of points of tangency of the isoprofit curves of the competitors) one or both firms would have higher profits, and hence industry profits would be higher. At point c firm B would retain the same profit (B_6) as at point e , while A would move to a higher profit level (A_9). At point d firm A would have the same profit (A_5) as at the Bertrand equilibrium e , but firm B would move to a higher isoprofit curve (B_{10}). Finally, at any point between c and d (e.g. at f) both firms would realise higher profits (A_7 and B_8) as compared to those attained at Bertrand's solution ($A_7 > A_5$ and $B_8 > B_6$).

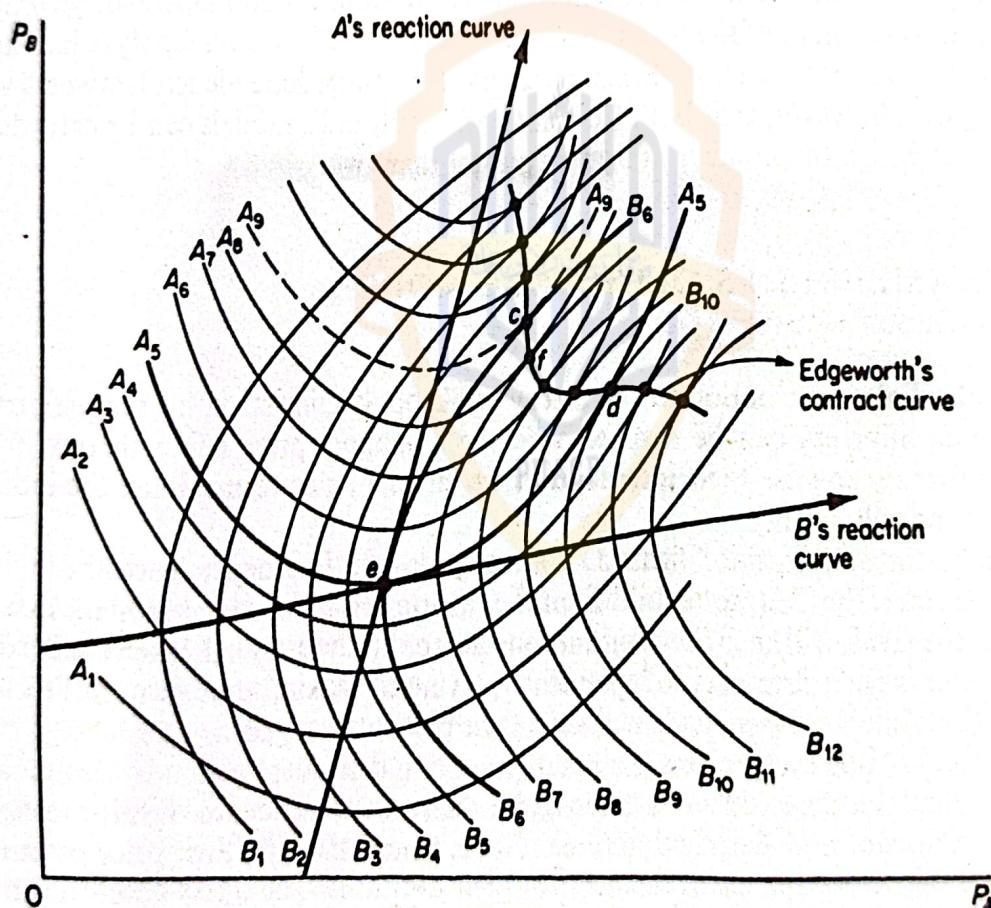


Figure 9.14

Bertrand's model may be criticised on the same grounds as Cournot's model:

The behavioural pattern emerging from Bertrand's assumption is naïve: firms never learn from past experience.

Each firm maximises its own profit, but the industry (joint) profits are not maximised.

The equilibrium price will be the competitive price. (In the example of costless mineral-water production, the price in Bertrand's model would fall to zero. If production is not costless, then price would fall to the level which would cover the costs of the duopolists inclusive of a normal profit.)

The model is 'closed' – does not allow entry.

The interesting feature of both Cournot's and Bertrand's models is that the limit of duopoly is pure competition. Neither model refutes the other. Each is consistent and is based on different behavioural assumptions. We may say that Bertrand's assumption (about the fixity of price of the rival) is more realistic, in view of the observed preoccupation of firms with keeping their prices constant (except in cost inflation situations). Furthermore, Bertrand's model focused attention on price setting as the main decision of the firm. The serious limitations of both models are the naïve behavioural pattern of rivals; the failure to deal with entry; the failure to incorporate other variables in the model, such as advertising and other selling activities, location of the plant, and changes in the product. Product differentiation and selling activities are the two main weapons of non-price competition, which is a main form of competition in the real business world; both models do not define the length of the adjustment process. Although dealing in terms of 'time periods,' their approach is basically static; both models assume that the market demand is known with accuracy; both models are based on individual demand curves which are located by making the convenient assumption of constant reaction curves of the competing firms.

Having discussed the classical duopoly models of Cournot and Bertrand, we proceed with the development of the traditional models of non-collusive oligopoly, which apply to market structures with a few firms conscious of their interdependence. It is worth while pointing out, however, that both Cournot's and Bertrand's models can be extended to markets in which the number of firms is greater than two.

V. STACKELBERG'S DUOPOLY MODEL

This model was developed by the German economist Heinrich von Stackelberg¹ and is an extension of Cournot's model. It is assumed, by von Stackelberg, that *one* duopolist is sufficiently sophisticated to recognise that his competitor acts on the Cournot assumption. This recognition allows the sophisticated duopolist to determine the reaction curve of his rival and incorporate it in his own profit function, which he then proceeds to maximise like a monopolist.

Assume that the isoprofit curves and the reaction functions of the duopolists are those depicted in figure 9.20. If firm *A* is the sophisticated oligopolist, it will assume that its rival will act on the basis of its own reaction curve. This recognition will permit firm *A* to choose to set its own output at the level which maximises its own profit. This is point *a* (in figure 9.20) which lies on the lowest possible isoprofit curve of *A*, denoting the maximum profit *A* can achieve given *B*'s reaction curve. Firm *A*, acting as a monopolist (by incorporating *B*'s reaction curve in his profit-maximising computations) will produce X_A , and firm *B* will react by producing X_B according to its reaction curve. The sophisticated oligopolist becomes in effect the leader, while the naïve rival who acts on the Cournot assumption becomes the follower. Clearly sophistication is rewarding for *A* because he reaches an isoprofit curve closer to his axis than if he behaved with the same naïveté as his rival. The naïve follower is worse off as compared with the Cournot equilibrium, since with this level of output he reaches an isoprofit curve further away from his axis.

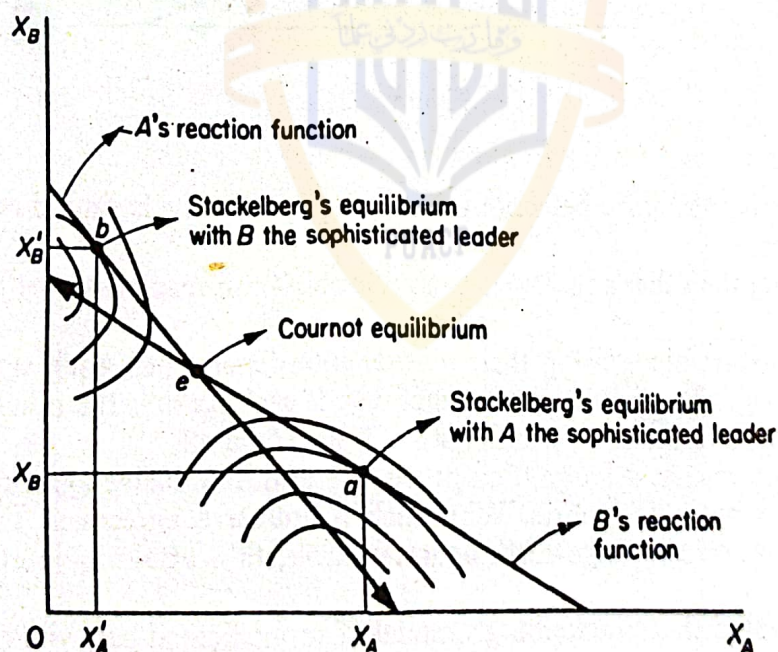


Figure 9.20

If firm *B* is the sophisticated oligopolist, it will choose to produce X'_B , corresponding to point *b* on *A*'s reaction curve, because this is the largest profit that *B* can achieve given his isoprofit map and *A*'s reaction curve. Firm *B* will now be the leader while firm *A* becomes the follower. *B* has a higher profit and the naïve firm *A* has a lower profit as compared with the Cournot equilibrium.

¹ H. von Stackelberg, *The Theory of the Market Economy*, trans. A. T. Peacock (London, 1952).

In summary, if only one firm is sophisticated, it will emerge as the leader, and a stable equilibrium will emerge, since the naïve firm will act as a follower.

However, if both firms are sophisticated, then both will want to act as leaders, because this action yields a greater profit to them. In this case the market situation becomes unstable. The situation is known as *Stackelberg's disequilibrium* and the effect will either be a price war until one of the firms surrenders and agrees to act as follower, or a *collusion* is reached, with both firms abandoning their naïve reaction functions and moving to a point closer to (or on) the Edgeworth contract curve with both of them attaining higher profits. If the final equilibrium lies on the Edgeworth contract curve the industry profits (joint profits) are maximised (figure 9.21).

Von Stackelberg's model has interesting implications.

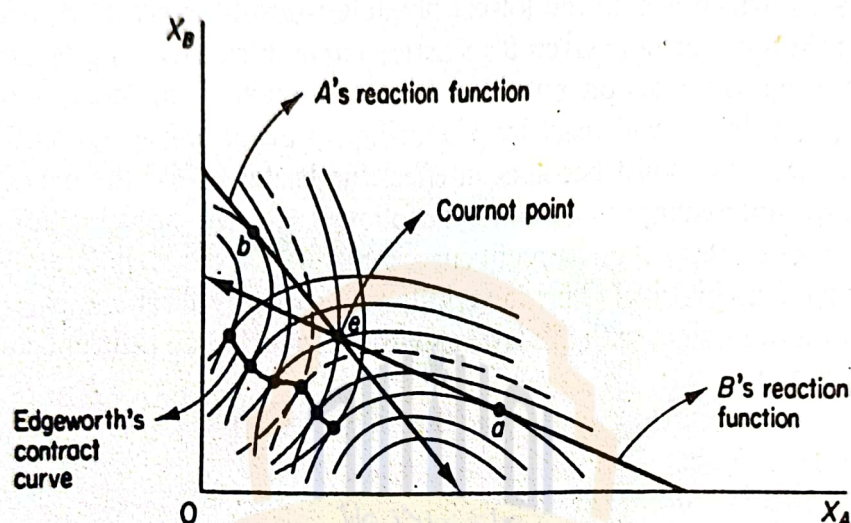


Figure 9.21

It shows clearly that naïve behaviour does not pay. The rivals should recognise their interdependence.

By recognising the other's reactions each duopolist can reach a higher level of profit for himself.

If both firms start recognising their mutual interdependence, each starts worrying about the rival's profits and the rival's reactions. If each ignores the other, a price war will be inevitable, as a result of which both will be worse off.

The model shows that a bargaining procedure and a collusive agreement becomes advantageous to both duopolists. With such a collusive agreement the duopolists may reach a point on the Edgeworth contract curve, thus attaining joint profit maximisation.

It should be noted that Stackelberg's model of sophisticated behaviour is not applicable in a market in which the firms behave on Bertrand's assumption. In a Cournot-type market the sophisticated firm 'bluffs' the rival, by producing a level of output larger than the one that would be produced in the Cournot equilibrium and the naïve rival, sticking to his Cournot behavioural reaction pattern, will be misled and produce less than in the Cournot equilibrium. However, in a Bertrand-type market the sophisticated duopolist can do nothing which would increase his own profit and persuade the other to stop price-cutting. The most he can do is to keep his own price constant, that is, behave exactly as his opponent expects him to behave.¹

¹ See Ira Horowitz, *Decision Making and the Theory of the Firm* (Holt, Rinehart & Winston, 1970) p. 180.

A numerical example of Stackelberg's model

Assume that in a duopoly market the demand function is

$$P = 100 - 0.5(X_1 + X_2)$$

and the duopolists' costs are

$$C_1 = 5X_1 \quad \text{and} \quad C_2 = 0.5X_2^2$$

The reaction functions are found by taking the partial derivatives of the duopolists' profit functions and equating them to zero:

$$\Pi_1 = PX_1 - C_1 = 95X_1 - 0.5X_1^2 - 0.5X_1X_2$$

$$\Pi_2 = PX_2 - C_2 = 100X_2 - X_2^2 - 0.5X_1X_2$$

The partial derivatives are

$$\frac{\partial \Pi_1}{\partial X_1} = 95 - X_1 - 0.5X_2 = 0$$

$$\frac{\partial \Pi_2}{\partial X_2} = 100 - 2X_2 - 0.5X_1 = 0$$

The reaction functions are

$$X_1 = 95 - 0.5X_2 \rightarrow A's \text{ reaction curve}$$

$$X_2 = 50 - 0.25X_1 \rightarrow B's \text{ reaction curve}$$

(1) Stackelberg's solution with A being the sophisticated leader

Firm A will substitute B's reaction function in its own profit equation, which it will then maximise as if it were a monopolist:

$$\Pi_1 = PX_1 - C_1 = 95X_1 - 0.5X_1^2 - 0.5X_1X_2$$

Substitute

$$X_2 = 50 - 0.25X_1$$

Maximise

$$\Pi_1 = 70X_1 - 0.375X_1^2$$

(a) First-order condition:

$$\frac{\partial \Pi_1}{\partial X_1} = 70 - 0.75X_1 = 0$$

This yields output:

$$X_1 = 93\frac{1}{3}$$

and profit:

$$\Pi_1 = 70X_1 - 0.375X_1^2 = 3267$$

(b) The second-order condition for profit maximisation is fulfilled.

Firm B would be the follower. It would assume that A would produce $93\frac{1}{3}$ units; thus B substitutes this amount in its reaction function

$$X_2 = 50 - 0.25X_1 = 26\frac{2}{3}$$

and its profit would be

$$\Pi_2 = 100X_2 - X_2^2 - 0.5X_1X_2 = 155.5$$

(2) Stackelberg's solution if firm B is the sophisticated duopolist

Firm B will substitute A's reaction function in its own profit function, and it will proceed to maximise this profit as a monopolist

$$\Pi_2 = PX_2 - C_2 = 100X_2 - X_2^2 - 0.5X_1X_2$$

Substitute $X_1 = 95 - 0.5X_2$ (i.e., A's reaction function)

$$\Pi_2 = 52.5X_2 - 0.75X_2^2$$

(a) The first-order condition for the maximisation of Π_2 requires

$$\frac{\partial \Pi_2}{\partial X_2} = 52.5 - 1.5X_2 = 0$$

which yields output:

$$X_2 = 35$$

and profit:

$$\Pi_2 = 52.5X_2 - 0.75X_2^2 = 918.75$$

(b) The second-order condition for the maximisation of Π_2 is fulfilled.

The follower is now firm A which will act on the Cournot assumption; it will assume that the rival will keep his quantity at $X_2 = 35$, and will find its own output by substituting this quantity in its reaction function.

$$X_1 = 95 - 0.5X_2 = 77.5$$

and its profit is

$$\Pi_1 = 95X_1 - 0.5X_1^2 - 0.5X_1X_2 = 3003$$

(3) Stackelberg's disequilibrium

If both entrepreneurs adopt Stackelberg's sophisticated pattern of behaviour, each will examine his profits if he acts as a leader and if he acts as a follower, and will adopt the action that will yield him the greatest profit.

Firm A calculates its profits both as a leader and as a follower:

If A is the leader his profits are 3267

If A is the follower his profits are 3003

Clearly firm A will prefer to act as the leader.

Firm B similarly, calculates its profits as a leader and as a follower:

If B is the leader his profits are 918.75

If B acts as the follower his profits are 155.50

Thus firm B will also choose to act as the leader.¹

With both firms acting in the sophisticated way implied by Stackelberg's behavioural hypothesis both will want to act as leaders. As they attempt to do so they find that their expectations about the rival are not fulfilled and 'warfare' will start, unless they decide to come to a collusive agreement.

We may now summarise Stackelberg's model. Each duopolist estimates the maximum profit that he would earn (a) if he acted as leader, (b) if he acted as follower, and chooses the behaviour which yields the largest maximum. Four situations may arise: (1) Duopolist A wants to be leader and B wants to be follower. (2) Duopolist B wants to be leader and A wants to be follower. (3) Both firms want to be followers. (4) Both firms desire to be leaders.

In situations (1) and (2) the result is a determinate equilibrium (provided that the first- and second-order conditions for maxima are fulfilled).

If both firms desire to be followers, their expectations do not materialise (since each assumes that the rival will act as a leader), and they must revise them. Two behavioural patterns are possible. If each duopolist recognises that his rival wants also to be a follower, the Cournot equilibrium is reached. Otherwise, one of the rivals must alter his behaviour and act as a leader before equilibrium is attained.

Finally, if both duopolists want to be leaders a disequilibrium arises, whose outcome, according to Stackelberg, is economic warfare. Equilibrium will be reached either by collusion, or after the 'weaker' firm is eliminated or succumbs to the leadership of the other.

¹ The numerical example is taken from J. M. Henderson and R. E. Quandt, *Microeconomic Theory* (McGraw-Hill 1958) p. 181.